



Psoriasis

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Psoriasis is an inflammatory skin disease that manifests most commonly as well-circumscribed, erythematous papules and plaques covered with silvery scales. Multiple factors contribute, including genetics. Common triggers include trauma, infection, and certain medications. Symptoms are usually minimal, but mild to severe itching may occur. Cosmetic implications may be major. Up to 30% of patients develop psoriatic arthritis. Diagnosis is based on appearance and distribution of lesions. Treatment can include topical agents (eg, glucocorticoids, vitamin D3 analogs, calcineurin inhibitors, tazarotene, roflumilast, tapinarof), phototherapy, and systemic medications (eg, [methotrexate](#), oral retinoids, [cyclosporine](#), and targeted biologic or oral small molecules).

Psoriasis causes hyperproliferation of epidermal keratinocytes combined with inflammation of the epidermis and dermis. It is also a chronic inflammatory disorder with potential multiorgan involvement.

Estimates of the prevalence of psoriasis vary widely across different populations, and it is likely that psoriasis is underreported among people with dark skin compared with people with light skin ([1](#)). In one systematic review of worldwide epidemiologic trends, psoriasis was found to be unequally distributed across geographical regions ([2,3](#)). The incidence of psoriasis was highest in Australasia (1.99%), followed by Western Europe (1.92%), Central Europe (1.83%), and North America (1.50%); it was lower in Southern Latin America (1.10%) and lowest in East Asia (0.14%). It is diagnosed more frequently in resource-rich countries and in regions with older populations. Adults are generally more commonly affected than children but the disorder can occur at any age. Some evidence suggests a rise in annual incidence rates, which may be partly attributed to evolving diagnostic standards ([4](#)).

General references

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Pathophysiology of Psoriasis

Psoriasis is a disease of keratinocyte hyperproliferation driven by inflammatory cascades of innate and adaptive immunity. Normal keratinocyte maturation and turnover takes approximately 28 days. However, in psoriasis this turnover is accelerated to 3 to 5 days because of the failure of keratinocyte maturation. The pathogenesis of psoriasis is driven by a complex interplay between genetic susceptibility, dysregulated immune responses, and additional environmental triggers (1). A family history of psoriasis is a strong risk factor. Certain genes and human leukocyte antigens (eg, HLA-Cw6, HLAB13, HLAB17) are associated with psoriasis. Genomewide linkage analysis has identified numerous psoriasis susceptibility loci; the *PSORS1* locus, within the major histocompatibility complex (MHC) on chromosome 6p21, plays the greatest role in determining a patient's susceptibility for developing psoriasis. Genetic defects in the MHC are associated with a wide range of autoimmune and primary immunodeficiency disorders, including psoriasis.

Alterations in both innate and adaptive immune responses also influence the disease pathogenesis. The central mechanism (usually genetically driven) involves activation of dendritic cells, which secrete interleukins (IL)-12 and IL-23, promoting differentiation and expansion of T-helper (Th)1, Th17, and Th22 cells. The most commonly implicated pathways are Th17-mediated, which produce IL-17 and IL-22. Keratinocytes producing IL-36 and Th1 cells producing interferon- γ and tumor necrosis factor- α (TNF- α), are also thought to contribute to pathogenesis, albeit to a lesser extent. An environmental trigger is thought to evoke further inflammatory responses and subsequent hyperproliferation of keratinocytes (2).

Well-identified triggers (risk factors) include:

- Injury (Koebner phenomenon)
- Sunburn
- Infections (viral, such as HIV; bacterial, such as beta-hemolytic streptococci associated with guttate psoriasis)
- Medications (especially beta-blockers, [chloroquine](#), [lithium](#), angiotensin-converting enzyme inhibitors, [indomethacin](#), [terbinafine](#), interferon- α , immune checkpoint inhibitors, and tumor necrosis factor inhibitors [3])
- Alcohol consumption
- Tobacco smoking
- Obesity

- Systemic glucocorticoid withdrawal (especially in erythrodermic or generalized pustular subtypes)

Experiencing psychological stress (4) has been associated with psoriasis but not conclusively established as a risk factor.

Complications of psoriasis include psoriatic arthritis and nail changes (such as pitting, leukonychia, nail crumbling, oil spots, subungual hyperkeratosis, and distal onycholysis) (1). Approximately 30% of patients develop psoriatic arthritis during their lifetime, and nail lesions are present in up to 90% of patients with psoriatic arthritis. Several disorders are thought to coexist (where evidence is insufficient to determine causality) with psoriasis (5). These include an increased risk of cardiovascular and metabolic diseases (eg, myocardial infarction, stroke, diabetes, obesity), anxiety, and depression.

Pathophysiology references

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Symptoms and Signs of Psoriasis

The classic morphology of psoriasis is well-demarcated, erythematous plaques with silvery scale. Lesions are either asymptomatic or pruritic and often localized on the scalp, extensor surfaces of the elbows and knees, sacrum, buttocks (commonly the gluteal cleft), and genitals. The nails and eyebrows may also be affected.

Plaques may appear violaceous in patients with dark skin and pink or red in patients with light skin, but key features such as scaling and distribution (eg, on extensor elbows, trunk, legs, and periumbilical area) are consistent across all skin tones.

Inverse psoriasis (also called flexural psoriasis) occurs in intertriginous areas including the axillary, umbilical, inframammary and perianal regions.

The disease can be widespread, involving confluent areas of skin extending between these regions.

Psoriasis is one of the most common underlying dermatoses leading to erythroderma (which is a clinical syndrome characterized by widespread erythema and scaling usually involving more than 75% of the skin), specifically in the form of erythrodermic psoriasis. Erythrodermic psoriasis is a rare and potentially life-threatening variant of psoriasis that presents with near total-body (75 to 90% body surface area)

generalized erythema and desquamation. The widespread inflammation may trigger symptoms like fevers, chills, malaise, and sequelae from large insensible fluid losses through the skin surface.

Morphology varies based on the subtype of psoriasis.

Manifestations of Psoriasis



Psoriasis Plaque

Psoriasis typically manifests as plaques covered with thick, silvery, shiny scales. Psoriasis plaques are elevated, palpable lesions > 10 mm in diameter.

Photo provided by Thomas Habif, MD.



Psoriasis (Plaques)

This image shows psoriatic plaques. Plaques may look violaceous in patients with dark skin tones, but key features such as scaling and distribution (eg, on extensor elbows, trunk, legs, and periumbilical area) are consistent across all skin tones.

Image courtesy of Karen McKoy, MD.



Psoriasis (Elbows)

This photo shows erythematous plaques covered with silvery-white scales on the extensor surface of the elbows. This appearance is typical of plaque psoriasis.

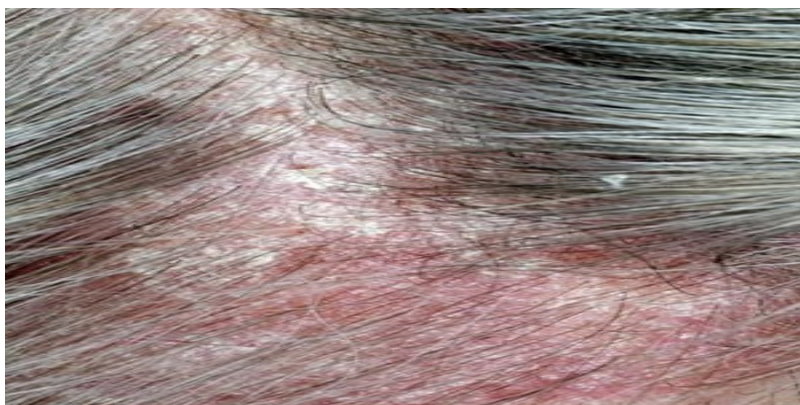
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Psoriasis (Fingers)

This photo shows patches of scaly, erythematous, and thick skin, and discolored, crumbly nails.

DR. HAROUT TANIELIAN/SCIENCE PHOTO LIBRARY



Psoriasis (Scalp)

This photo shows red, flaky, and thickened skin on the scalp of a patient with psoriasis.

DR P. MARAZZI/SCIENCE PHOTO LIBRARY

Nail Psoriasis With Pitting and Discoloration

This photo shows irregular pits and areas of yellow-brown discoloration (oil spots) in a person with psoriasis.

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Nail Psoriasis With Thickening and Crumbling of the Nail Plate

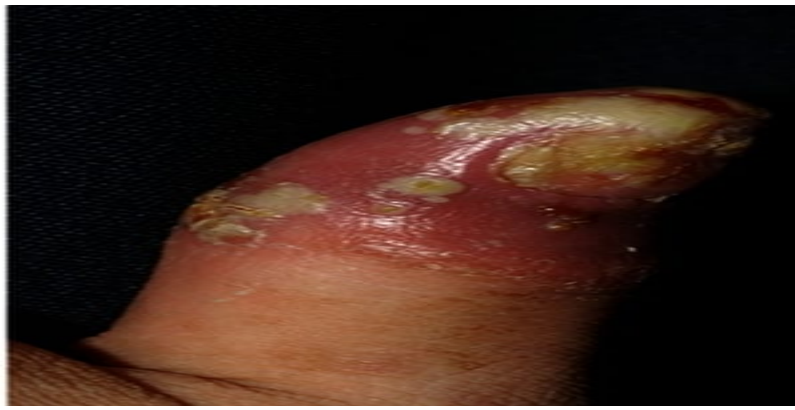
The photo on the left shows thickening and crumbling of the nail of the great toe. Pitting and onycholysis are visible on the hand (right), suggesting nail psoriasis as the diagnosis.

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Pustular Psoriasis (Feet)

This image shows widespread pustules on the soles of a patient with erythrodermic pustular psoriasis.

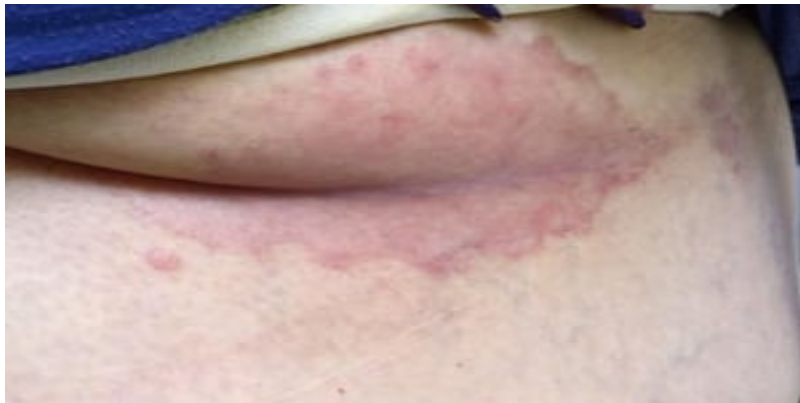
Image courtesy of Karen McKoy, MD.



Acrodermatitis Continua of Hallopeau

This photo shows pustular lesions on the tip of a finger in a patient with acrodermatitis continua of Hallopeau (a localized form of pustular psoriasis).

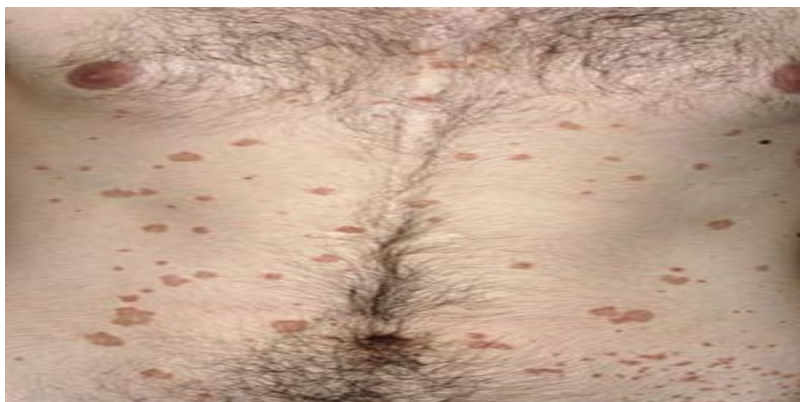
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Inverse Psoriasis (Flexural psoriasis)

In inverse psoriasis, red, shiny patches (sometimes without scales) form in the intertriginous areas such as the inframammary fold.

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Guttate Psoriasis

This photo shows multiple plaques on the trunk of a 22-year-old man with guttate psoriasis.

DR P. MARAZZI/SCIENCE PHOTO LIBRARY



Erythrodermic Psoriasis

This photo shows a diffuse erythematous rash on the torso of a patient with erythrodermic psoriasis.

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Palmoplantar Psoriasis (Palm)

This photo shows hyperkeratotic discrete plaques on the palm of a patient with palmoplantar psoriasis.

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Palmoplantar Psoriasis (Sole)

This photo shows hyperkeratotic discrete plaques on the sole of a patient with palmoplantar psoriasis.

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Palmoplantar Pustulosis

This photo shows erythematous macules and pustules on the foot of a patient with pustular psoriasis.

Image courtesy of Karen McKoy, MD.

Among the various psoriasis subtypes (see table [Subtypes of Psoriasis](#)), plaque psoriasis (psoriasis vulgaris or chronic plaque psoriasis) accounts for approximately 90%; lesions are discrete, erythematous papules or plaques covered with thick, silvery, shiny scales. Lesions appear gradually and remit and recur spontaneously or with the appearance and resolution of triggers.

Psoriatic arthritis can be disabling ([psoriatic arthritis](#)); joint destruction may ultimately occur.

Psoriasis is rarely life threatening but can affect a patient's self-image. Besides the patient's appearance, the sheer amount of time required to treat extensive skin or scalp lesions with topical agents and to maintain clothing and bedding may adversely affect quality of life.

TABLE

Subtypes of Psoriasis

Subtype	Description	Treatment and Prognosis
Acrodermatitis continua of Hallopeau	Pustular psoriasis confined to distal fingers or toes, sometimes to just one digit Replaced by scale and crust when it resolves	Treatment: Systemic retinoids, vitamin D3 analogs (eg, calcipotriol), topical glucocorticoids, systemic immunosuppressants (eg, biologics, small molecules) Prognosis: Waxes and wanes
Erythrodermic psoriasis	Gradual or sudden onset of diffuse erythema, usually in patients with	Treatment: Systemic immunosuppressants (eg, methotrexate ,

Subtype	Description	Treatment and Prognosis
	plaque psoriasis (possibly the first manifestation of erythrodermic psoriasis); typical psoriatic plaques less prominent or absent Most commonly triggered by inappropriate use of topical or systemic glucocorticoids or light therapy	cyclosporine, biologics, small molecules) or intense topical therapy, sometimes as inpatient therapy Tars, anthralin, and phototherapy likely to exacerbate the condition Prognosis: Good with elimination of triggering factors
Generalized pustular psoriasis	Explosive onset of widespread erythema and sterile pustules	Treatment: Systemic immunosuppressants (eg, biologics, cyclosporine) Prognosis: If untreated, can be fatal due to high-output heart failure
Guttate psoriasis	Abrupt appearance of multiple plaques 0.5 to 1.5 cm in diameter, usually on the trunk in children and young adults after streptococcal pharyngitis	Treatment: Antibiotics for underlying streptococcal infection as needed Possibly topical and/or systemic treatment after antibiotics Prognosis: Excellent, often with permanent cure May progress to plaque psoriasis
Inverse psoriasis	Psoriasis of intertriginous areas (usually the inguinal, gluteal, axillary, inframammary, and retroauricular folds and the glans of the uncircumcised penis) Possibly formation of cracks or fissures in the	Treatment: Topical glucocorticoids of minimal effective potency, with or without vitamin D3 analogs (eg, calcipotriol) Possibly tacrolimus 0.1% ointment in recalcitrant cases

Subtype	Description	Treatment and Prognosis
	center or edge of involved areas Possibly absence of scales	Tar and anthralin possibly irritating Prognosis: Waxes and wanes
Nail psoriasis	Pitting, stippling, fraying, discoloration (oil spot sign), and thickening of the nails, with or without separation of the nail plate (onycholysis) May resemble a fungal nail infection Affects 30–50% of patients with other forms of psoriasis	Treatment: Responds best to systemic therapy with immunosuppressants (eg, biologics, small molecules) For brave or stoic patients, possibly intralesional injection of nail folds with glucocorticoids Prognosis: Often unresponsive to treatment
Palmoplantar psoriasis	Hyperkeratotic, discrete plaques on palms and/or soles that tend to become confluent	Treatment: Systemic retinoids, topical glucocorticoids, vitamin D3 analogs (eg, calcipotriol), systemic immunosuppressants (eg, methotrexate , cyclosporine , biologics, small molecules) Prognosis: Waxes and wanes Rarely resolves completely, even with treatment
Plaque psoriasis	Gradual appearance of discrete, erythematous papules or plaques covered with thick, silvery, shiny scales Lesions that remit and recur spontaneously or	Treatment: Topical glucocorticoids of minimal effective potency, with or without vitamin D3 analogs (eg, calcipotriol) Systemic immunosuppressants (eg,

Subtype	Description	Treatment and Prognosis
	with appearance and resolution of triggers	methotrexate , cyclosporine , biologics, small molecules) Prognosis: Waxes and wanes
Palmoplantar pustulosis (pustular psoriasis of the palms and soles)	Gradual appearance of deep pustules on palms and soles Flares may be painful and disabling Typical psoriatic lesions possibly absent	Treatment: Systemic retinoids or psoralen plus ultraviolet A (PUVA) therapy Prognosis: Waxes and wanes

Diagnosis of Psoriasis

- Primarily physical examination
- Rarely biopsy

The diagnosis of psoriasis is most often based on clinical appearance and distribution of lesions.

Differential diagnoses include:

- [Seborrheic dermatitis](#)
- [Dermatophytoses](#) (potassium hydroxide wet mount should be performed for any scaly plaques, especially if they do not have a classic appearance of eczema or psoriasis)
- [Cutaneous lupus erythematosus](#)
- [Eczema](#)
- [Lichen planus](#)
- [Allergic contact dermatitis](#)
- [Pityriasis rosea](#)
- Squamous cell carcinoma in situ ([Bowen disease](#), especially when on the trunk; this diagnosis should be considered for isolated plaques that do not respond to usual therapy)
- [Lichen simplex chronicus](#)
- [Secondary syphilis](#)

Biopsy is rarely necessary and may not be diagnostic; however, it may be considered in cases where the clinical findings are not classic or the patient is not responding as expected to standard therapies.

Disease is graded classically as mild, moderate, or severe based on the body surface area affected and how the lesions affect the patient's quality of life. Beyond body surface area, psoriasis severity is also assessed by considering involvement of high-impact body sites (eg, face, hands) when determining candidacy for topical therapy versus phototherapy or systemic treatments.

Treatment of Psoriasis

- Topical treatments
- Phototherapy [Ultraviolet (UV) light therapy]
- Systemic therapies

Treatment options are extensive and range from topical treatments (eg, glucocorticoids, vitamin D3 analogs, calcineurin inhibitors, [tazarotene](#), [roflumilast](#), [tapinarof](#), emollients, [salicylic acid](#), [coal tar](#)) to phototherapy (eg, UV light therapy) to systemic treatments (eg, [methotrexate](#), oral retinoids, [cyclosporine](#), biologics, small molecules).

The primary consideration in clinical decision-making guiding the initial choice of treatment is whether a psoriasis patient has concurrent psoriatic arthritis, in which case systemic therapy should be considered. If psoriatic arthritis is not present, then treatment decisions are tailored based on factors such as body surface area, involvement of high impact areas, patient preference, quality of life considerations, prior response to topical agents, and the presence of other comorbidities.

Choice of therapy

The choice of specific agents and combinations requires close cooperation with the patient, always keeping in mind the adverse effect profile. There is no single ideal combination or sequence of agents, but treatment should be kept as simple as possible. Monotherapy is preferred, but combination therapy is the norm.

Rotational therapy refers to the substitution of one therapy for another after 1 to 2 years to reduce the adverse effects caused by chronic use and to circumvent disease resistance. Sequential therapy refers to initial use of potent agents (eg, [cyclosporine](#)) to quickly gain control followed by use of agents with a better safety profile. Immunosuppressants achieve clearance or near clearance of lesions more often than [methotrexate](#) or narrowband UVB.

Mild plaque psoriasis can be treated with emollients, keratolytics, tar, topical glucocorticoids, glucocorticoid-sparing topical agents (eg, calcineurin inhibitors, PDE-4 inhibitors, [tapinarof](#)), vitamin D3 analogs, or [anthralin](#) alone or in combination. Moderate exposure to sunlight is beneficial, but sunburn can induce exacerbations.

Moderate to severe plaque psoriasis should be treated with topical agents and either phototherapy or systemic agents. Immunosuppressants are used for moderate to severe disease unresponsive to other agents.

Scalp plaques are often difficult to treat because of the physical challenge of applying topical agents and shielding skin from UV light. Scalp disease can be particularly burdensome to patients. Multiple formulations that are more friendly to scalp application include alcohol- or water-based solutions and liquids, foams, shampoos, gels and, in more recalcitrant cases, oils (that are applied overnight and washed off in the morning). A systematic review of randomized trials found that several IL-17 inhibitors, including [brodalumab](#), [secukinumab](#), and [ixekizumab](#) (in a subgroup of patients with severe psoriasis), show high efficacy in treating moderate to severe scalp psoriasis, with [brodalumab](#) and [ixekizumab](#) demonstrating a rapid response (1). [Guselkumab](#) (IL-23 inhibitor) is superior to [adalimumab](#) (TNF inhibitor), and [ixekizumab](#) is superior to [etanercept](#) (TNF inhibitor). [Apremilast](#) offers long-term efficacy.

Moderate to severe cases of scalp psoriasis may benefit from treatment with systemic immunosuppressants (eg, biologics, small molecules).

Special treatment needs for [subtypes of psoriasis](#) are described above.

For [psoriatic arthritis](#), treatment with systemic therapy is important to consider in order to prevent joint destruction, decrease pain, and improve or retain function.

(See also the American Academy of Dermatology Association's [Psoriasis Clinical Guideline](#).)

Topical treatments

Glucocorticoids are usually used topically but can be injected into small or recalcitrant plaques.

(CAUTION: *Systemic glucocorticoids may precipitate exacerbations or development of pustular psoriasis and should not be used to treat psoriasis.*) Topical glucocorticoids are the mainstay for mild to mild-moderate psoriasis, with potency and formulation appropriately tailored to the site and extent of involvement. Topical glucocorticoids are used twice daily. Glucocorticoids are most effective when used overnight under occlusive polyethylene coverings or incorporated into tape; a glucocorticoid cream is applied without occlusion during the day. [Glucocorticoid potency](#) is selected according to the extent of involvement. Topical glucocorticoid use may be limited by logistical considerations (eg, cost, inconvenience when treating extensive areas) and potential for adverse effects. Topical glucocorticoids applied for long periods to large areas of the body may also cause systemic effects and paradoxically exacerbate psoriasis.

As lesions abate, the glucocorticoid should be applied less frequently or at a lower potency to minimize local atrophy, striae formation, and telangiectases. Ideally, after approximately 2 to 3 weeks, glucocorticoids should be substituted with a glucocorticoid-sparing agent.

For chronic or maintenance therapy, regimens (eg, twice weekly application) with both topical glucocorticoids, and glucocorticoid-sparing alternatives such as topical [tapinarof](#) or [roflumilast](#) are recommended to reduce the risk of skin atrophy and maintain remission. [Vitamin D](#) analogs are an alternative. Phototherapy and systemic agents should also be considered. If glucocorticoid-sparing options are effective alone, monotherapy may be acceptable, especially for areas more susceptible to glucocorticoid-induced side effects. This substitution limits topical glucocorticoid exposure and reduces risk of adverse effects (eg, skin atrophy, telangiectasias, easy bruising, striae). For small, thick, localized, or recalcitrant lesions, high-potency glucocorticoids are used with an occlusive dressing or

[flurandrenolide](#) tape; these dressings are left on overnight and changed in the morning. Relapses after topical glucocorticoid therapy is stopped often occur faster than with other agents.

Vitamin D3 analogs (eg, calcipotriol [[calcipotriene](#)], [calcitriol](#)) are topical medications that induce normal keratinocyte proliferation and differentiation; they can be used alone or in combination with topical glucocorticoids. Some clinicians have patients apply calcipotriol on weekdays and glucocorticoids on weekends.

Calcineurin inhibitors (eg, [tacrolimus](#), [pimecrolimus](#)) are available in topical form and are generally well-tolerated. They are not as effective as glucocorticoids but may avoid the complications of glucocorticoids when treating facial and intertriginous psoriasis. Data from large cohort studies and meta-analyses evaluating the risk of lymphoma associated with topical calcineurin inhibitor use are mixed. Some studies have reported a small but statistically significant increase in lymphoma risk ([2](#)), while others have found no increased risk ([3](#)). In addition, the overall evidence does not support an increase in skin cancer risk.

Tazarotene is a topical retinoid. It is less effective than glucocorticoids as monotherapy but is a useful adjunct because of its keratolytic effect. It is typically used on thicker psoriasis plaques that may appear on the trunk and extremities but also may be used on the face.

Roflumilast is a topical phosphodiesterase-4 (PDE-4) inhibitor. It exerts anti-inflammatory and antipruritic effects by downregulating proinflammatory cytokines and normalizing keratinocyte function. It is available in cream and foam formulations. [Roflumilast](#) cream is particularly useful as a glucocorticoid-sparing treatment for delicate facial and intertriginous areas (eg, buttocks, groin, axillae). The foam formulation is useful in penetrating lesions in the scalp and/or other hirsute (hair-covered) areas. The most common side effect of topical [roflumilast](#) use is headache.

Tapinarof is a topical aryl hydrocarbon receptor (AhR) agonist used as monotherapy or in combination with topical glucocorticoids or phototherapy for the treatment of mild to moderate psoriasis. It is also useful as a glucocorticoid-sparing therapy on facial and intertriginous areas. Common adverse effects include irritant erythema, burning, and pruritus at the site of application, especially when used in intertriginous areas; acne and folliculitis have also been described.

Other adjunctive topical treatments include emollients, [salicylic acid](#), [coal tar](#), and [anthralin](#):

- **Emollients** include emollient creams, ointments, petrolatum, paraffin, and even hydrogenated vegetable (eg, coconut) oils. They reduce scaling and are most effective when applied twice daily and immediately after bathing. Lesions may appear redder as scaling decreases or becomes more transparent. Emollients are safe and should generally be used as an adjunct for mild to moderate plaque psoriasis. While emollient creams and ointments do not provide disease-modification, they enhance skin barrier repair and reduce scaling and/or dryness, which can improve appearance and comfort. Thicker emollients (like petrolatum) can enhance the penetration of topical active agents, and [mineral oil](#) can enhance UV penetration if coadministered with phototherapy.

- **Salicylic acid** is a keratolytic that softens scales, facilitates their removal, and increases absorption of other topical agents. It is especially useful as a component of scalp treatments; scalp scale can be quite thick.
- **Coal tar** preparations are anti-inflammatory and decrease keratinocyte hyperproliferation via an unknown mechanism. Ointments or solutions are typically applied at night and washed off in the morning. Coal tar products can be used in combination with topical glucocorticoids or with exposure to natural or artificial broad-band UVB light (280 to 320 nm) in slowly increasing increments (Goeckerman regimen). Shampoos should be left in for 5 to 10 minutes and then rinsed out.
- **Anthralin** (dithranol) is a topical antiproliferative, anti-inflammatory agent that is not a preferred treatment in geographical regions where other agents are available. Its mechanism of action is unknown. Effective dose is 0.1% cream or ointment increased to 1% as tolerated. Anthralin may be irritating and should be used with caution in intertriginous areas; it also stains. Irritation and staining can be avoided by washing off the anthralin 20 to 30 minutes after application. Using a liposome-encapsulated preparation may also avoid some disadvantages of anthralin.

Given the availability and convenience of other agents, **coal tar** and **anthralin** are being used less frequently in resource-rich settings where alternative therapies are available.

Phototherapy

UV light therapy is typically used in patients with extensive psoriasis; however, its use is declining because of the availability of effective alternative systemic therapies. The mechanism of action is poorly characterized; however, UVB light is known to reduce DNA synthesis and can induce mild systemic immunosuppression. Narrowband UVB light (311 to 312 nm), which is used without psoralens, is similar in effectiveness to psoralen plus ultraviolet A (PUVA). However, in settings where narrow-band UVB is available, that is clinically preferred to PUVA. One study has demonstrated that home-based narrow-band UVB phototherapy for plaque or guttate psoriasis is as effective as and less burdensome than office-based therapy (4).

In PUVA, oral methoxypsoralen, a photosensitizer, is followed by exposure to long-wave UVA light (330 to 360 nm). PUVA has an antiproliferative effect and also helps to normalize keratinocyte differentiation. Doses of light are started low and increased as tolerated. However, adverse effects limit its use. Severe burns can result if the dose of medication or UVA is too high.

Although the treatment is less messy than topical treatments and may produce remissions lasting several months, repeated treatments may increase the incidence of UV-induced skin cancer and melanoma. Less UV light is required when used with oral retinoids (the so-called re-PUVA regimen). Excimer laser therapy is a type of phototherapy using a 308-nm laser directed at focal psoriatic plaques

Systemic therapies

Methotrexate taken orally is a modestly effective treatment for moderate to severe psoriasis with mixed efficacy data for psoriatic arthritis. **Methotrexate** may also be used in some cases of pustular

psoriasis and palmoplantar disease unresponsive to topical agents or UV light therapy (narrowband UVB). Hematologic, renal, and hepatic function should be monitored. Dosage regimens vary, so only clinicians experienced in its use for psoriasis should undertake methotrexate therapy.

Cyclosporine can be used for severe psoriasis. It should be limited to courses of several months (rarely, up to 1 year) and alternated with other therapies. Its effect on the kidneys and potential long-term effects on the immune system preclude more liberal use.

Systemic retinoids (eg, [acitretin](#), [isotretinoin](#)) may be effective for severe and recalcitrant cases of psoriasis vulgaris, pustular psoriasis (in which isotretinoin may or may not be preferred), and hyperkeratotic palmoplantar psoriasis. Because of the teratogenic potential and long-term retention of [acitretin](#) in the body, it should not be used in pregnant patients or those with child bearing potential; such patients should also be warned against becoming pregnant for at least 3 years after treatment ends. Pregnancy restrictions also apply to isotretinoin, but the agent is not retained in the body beyond 1 month. Long-term treatment may be associated with [diffuse idiopathic skeletal hyperostosis](#) (DISH).

Several classes of **biologics** (therapeutic monoclonal antibodies) have shown efficacy for the treatment of moderate to severe plaque psoriasis ([5, 6](#)):

- **Tumor necrosis factor (TNF) inhibitors:** [Etanercept](#), [adalimumab](#), [infliximab](#), [golimumab](#) (only for psoriatic arthritis) and [certolizumab pegol](#) (does not cross the placenta)
- **Interleukin (IL)-23 inhibitors:** [Tildrakizumab](#), [risankizumab](#), [guselkumab](#), and [ustekinumab](#) (targets both IL-12 and IL-23)
- **IL-17 inhibitors:** [Secukinumab](#), [ixekizumab](#), [bimekizumab](#), and [brodalumab](#)

Apremilast, a PDE-4 inhibitor, is an oral small-molecule medication for plaque psoriasis; however, data suggest it is less effective than biologics ([7](#)).

Deucravacitinib is an oral small-molecule agent that inhibits tyrosine kinase 2 and has shown efficacy for treatment of psoriasis ([8, 9](#)).

Janus kinase inhibitors (eg, [tofacitinib](#), [upadacitinib](#)) are used to treat psoriatic arthritis, and also appear to be effective for psoriasis ([10](#)); however, more studies are needed before they are routinely used for this indication.

Spesolimab is an IL-36 inhibitor that is used for the treatment of generalized pustular psoriasis.

Biosimilar medications for many of the biologic classes above are available or under development ([11](#)). These medications have very similar efficacies and side effect profiles to reference products; slight differences in components do not appear to be clinically meaningful.

Other immunosuppressants (eg, [hydroxyurea](#), [6-thioguanine](#)) have narrow safety margins and are reserved for severe, recalcitrant psoriasis. Efalizumab is not available in the United States or Canada because of increased risk of [progressive multifocal leukoencephalopathy](#).

Emerging therapeutics for psoriasis include targeted oral peptides. [Icotrokinra](#), which is an IL-23 inhibitor, has shown efficacy and a tolerable safety profile in randomized trials of adults with moderate-

to-severe psoriasis ([12](#), [13](#)).

Fumaric acid esters (eg, [dimethyl fumarate](#), monoethyl fumarate) are compounds with immunomodulatory properties sometimes used for the treatment of moderate to severe plaque psoriasis, particularly in Europe, where they are a common first-line systemic therapy ([14](#)).

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Guidelines for Psoriasis

Following are professional medical society or government clinical practice guidelines regarding this medical issue; this is not a comprehensive list:

- [American Academy of Dermatology Association](#): Psoriasis Clinical Guideline; Joint guideline from the American Academy of Dermatology and the National Psoriasis Foundation
- [National Psoriasis Foundation Treatment Guidelines](#): National Psoriasis Foundation and the American College of Rheumatology joint treatment guideline for psoriatic arthritis
- [PsOPsA Hub](#): Updated Group for Research and Assessment of Psoriasis and Psoriatic Arthritis (GRAPPA) treatment guidelines for psoriatic arthritis

Key Points

- Psoriasis is a common inflammatory disorder affecting the skin that has a genetic component and several environmental triggers (eg, trauma, infection, certain medications).
- The most common skin findings are usually well-circumscribed, erythematous papules and plaques covered with silvery scales in plaque psoriasis, but lesions differ between the other less common subtypes of psoriasis.
- Psoriatic arthritis develops in 5 to 30% of patients and can cause joint destruction and disability.
- Diagnose based on the appearance and distribution of lesions.
- Topical treatments (eg, emollients, [salicylic acid](#), [coal tar](#) preparations, [anthralin](#), glucocorticoids, vitamin D3 analogs, calcineurin inhibitors, PDE-4 inhibitors, [tapinarof](#), [tazarotene](#)) are recommended for mild disease.
- For moderate to severe psoriasis, systemic treatments are considered, such as retinoids, [methotrexate](#), [cyclosporine](#), biologics (eg, tumor necrosis factor inhibitors), and small molecules (eg, [deucravacitinib](#), [apremilast](#)).

- Ultraviolet (UV) light therapy may be an alternative for patients who wish to avoid systemic immunosuppression.

Drug Information for the Topic



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